

Lecture 3 Module 2

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Countable & Uncountable Sets

- X countable : $X \leq \omega$
- X countably infinite : $X \sim \omega$
- X uncountable : $\omega < X$.

Obs. • Every subset of ω is countable.
 • Every subset of a countable set is countable.

Consequences

1. If $f: \omega \rightarrow X$ is onto, X is countable.
 - To show $X \leq \omega$.
 - for any $x \in X$, $f^{-1}(\{x\}) \neq \emptyset$.
 $(\subseteq \omega)$
 - Define $g: X \rightarrow \omega$
 s.t. $f(g(x)) = x$.
 - g is one-one.

2. X is countable iff there is a countable set Y and $f: Y \rightarrow X$ s.t. f is onto.

3. Y is countably infinite.
 $X (\neq \emptyset)$ is countable iff there is $f: Y \rightarrow X$ that is onto.

$(\Leftarrow)$: $Y \sim \omega$.. use 1:

$(\Rightarrow)$: $X \leq_{\mathfrak{g}} \omega$ - so get $f: \omega \rightarrow X$ that is onto.



4. Union of a pair of countable sets is countable.

- A, B : countably infinite
 $f: A \rightarrow X$, $g: B \rightarrow Y$ onto

→ Unions of finite family / countable family /

Countably infinite family
of countable sets : countable.

Qs. Is every set countable?

— A^B : set of all functions from
 B to A

For finite sets, $|A^B| = |A|^{|B|}$.

— $\mathbb{2} := \{0, 1\}$. $\mathcal{P}(A) \sim \mathbb{2}^A$

$X \subseteq A$. $\chi_X: A \rightarrow \mathbb{2}$

$$\chi_X(a) = \begin{cases} 1 & \text{if } a \in X \\ 0 & \text{if } a \in A \setminus X. \end{cases}$$

χ_X : Characteristic function of X

→ For finite A , $|\mathcal{P}(A)| = 2^{|A|}$.

Cantor's Theorem

Every set is strictly dominated
by its power set: $X < \mathcal{P}(X)$.

Proof: $X \leq \mathcal{P}(X)$

• Suppose $X \not\sim \mathcal{P}(X)$.

- Consider $Y := \{a \in X : a \notin f(a)\}$
 $\in \mathcal{P}(X)$.

- There is $a \in X$ s.t. $f(a) = Y$.

Q5. $a \in Y$

$a \notin Y$

$\mathcal{P}(\omega)$: uncountable.

- Continuum hypothesis: